

AFCNNNet: Automated detection of AF using chirplet transform and deep convolutional bidirectional long short term memory network with ECG signals

Tejas Radhakrishnan^{a,1}, Jay Karhade^{a,1}, S.K. Ghosh^a, P.R. Muduli^b, R.K. Tripathy^{a,*},
U. Rajendra Acharya^{c,d,e}

^a Department of Electrical and Electronics Engineering, BITS-Pilani, Hyderabad Campus, Hyderabad, 500078, India

^b Department of Electronics Engineering, Indian Institute of Technology (BHU), Varanasi, Uttar Pradesh, 221005, India

^c Department of Electronics and Computer Engineering, Ngee Ann Polytechnic, Singapore

^d Department of Bioinformatics and Medical Engineering, Asia University, Taichung, Taiwan

^e Department of Biomedical Engineering, School of Science and Technology, SUSS University, Singapore

ARTICLE INFO

Keywords:

Atrial fibrillation
ECG signal
Chirplet transform
Deep CNN-BLSTM
Classification
Smart healthcare

ABSTRACT

Atrial fibrillation (AF) is the most common type of cardiac arrhythmia and is characterized by the heart's beating in an uncoordinated manner. In clinical studies, patients often do not have visible symptoms during AF, and hence it is harder to detect this cardiac ailment. Therefore, automated detection of AF using the electrocardiogram (ECG) signals can reduce the risk of stroke, coronary artery disease, and other cardiovascular complications. In this paper, a novel time-frequency domain deep learning-based approach is proposed to detect AF and classify terminating and non-terminating AF episodes using ECG signals. This approach involves evaluating the time-frequency representation (TFR) of ECG signals using the chirplet transform. The two-dimensional (2D) deep convolutional bidirectional long short-term memory (BLSTM) neural network model is used to detect and classify AF episodes using the time-frequency images of ECG signals. The proposed TFR based 2D deep learning approach is evaluated using the ECG signals from three public databases. Our developed approach has obtained an accuracy, sensitivity, and specificity of 99.18% (Confidence interval (CI) as [98.86, 99.49]), 99.17% (CI as [98.85, 99.49]), and 99.18% (CI as [98.86, 99.49]), respectively, with 10-fold cross-validation (CV) technique to detect AF automatically. The proposed approach also classified terminating and non-terminating AF episodes with an average accuracy of 75.86%. The average accuracy value obtained using the proposed approach is higher than the short-time Fourier transform (STFT), discrete-time continuous wavelet transform (DT-CWT), and Stockwell transform (ST) based time-frequency analysis methods with deep convolutional BLSTM models to detect AF. The proposed approach has better AF detection performance than the existing deep learning-based techniques using ECG signals from the MIT-BIH database.

1. Introduction

Cardiac ailments are the leading causes of high mortality rates globally, and the primary forms of these ailments are coronary artery disease (CAD), and stroke [1]. Atrial fibrillation (AF) is a type of cardiac ailment that, if left untreated, can increase the risk of various other life-threatening cardiac diseases [2]. It is the most common sustained cardiac ailment, occurring in 1%–2% of the general population [3]. The prevalence of AF increases with improper diet and age. AF pathology

affects 5–15% of people over 80 and less than 0.5% of people over 40–50 years [4]. Since most episodes of AF pathology are asymptomatic [3,4], relying on physical symptoms to detect AF is not always feasible. Recently, the study in Ref. [5] found that chronic obstructive pulmonary disease (COPD) increases the risk of developing AF pathology. The AF is categorized into terminating AF (paroxysmal AF) and non-terminating AF (sustained AF) [6]. The AF episodes that terminate immediately are interpreted as paroxysmal AF. AF is also characterized based on the occurrences of multiple wavelets in the atria [7,8]. The termination of

* Corresponding author. Department of EEE, BITS Pilani, Hyderabad, India.

E-mail addresses: rajeshitg13@gmail.com, rajesh.nitr11@gmail.com (R.K. Tripathy).

¹ Both authors have equal contributions in this paper.

paroxysmal AF episodes depends on the number of circulating wavelets produced in the atria [7]. The non-terminating or sustained AF contains more circulating wavelets in the atria as compared to the paroxysmal AF [9]. The sustained AF is the primary reason for the development of thrombus, which leads to myocardial infarction (MI) and sudden cardiac death (SCD) [9,10]. Therefore, the early detection of AF and classification of paroxysmal vs. sustained AF episodes are essential in the clinical study to minimize the chances of heart attack and SCD. The Electrocardiogram (ECG) test is considered as the primary diagnostic procedure for the detection of AF pathology in the clinical standard [11]. The echocardiography test is also performed to diagnose any other cardiac complications such as conduction defects, valve defects, and coronary artery diseases due to AF pathology [12]. The AF symptoms in the ECG signal are the presence of irregular QRS-complexes, irregular and chaotic F-waves instead of normal P-waves, and the variations in the RR-intervals, respectively. In clinical studies, the 24/48-h Holter monitoring of ECG recording is helpful to detect more paroxysmal AF episodes [13]. The ambulatory ECG monitoring has the advantages of recording the heart electrical activity for longer durations, and it has reported better diagnostic performance for the detection of AF and other cardiac ailments as compared to the 24/48-h Holter monitoring [14]. A huge amount of ECG data are produced during ambulatory and 24/48-h Holter-based ECG monitoring for the diagnosis of AF disease [15]. The manual interpretation of AF pathology using this huge volume of ECG data is a cumbersome process. Therefore, the intelligent healthcare systems are used for the automated diagnosis of AF for ambulatory and longer duration Holter ECG monitoring systems [16,17]. The intelligent healthcare systems mainly consist of four essential steps such as the recording of ECG sensor data, extraction of different features from the ECG data using signal processing techniques, selection of relevant features, and detection and classification of AF episodes using different machine learning approaches [15,16]. The performance of the intelligent healthcare system depends on the selected features and the methods used for the evaluation features. An alternative to overcome this issue is to use deep learning techniques, in which the whole ECG signals are used as input to the model. By doing this, steps such as the evaluation of features, selection of features, and classification are not required in deep learning-based intelligent healthcare systems [18,19]. The development of new automated deep neural network-based algorithms for the detection of AF and classification of paroxysmal (terminating) AF and sustained (non-terminating) AF episodes using single-channel ECG signals is an interesting and active research topic for the design of smart healthcare systems.

In the last two decades, various automated algorithms have been proposed to detect AF using ECG signals [20–22]. These existing algorithms have used analysis of atrial activity (AA) and heartbeat durations (HBD) signals to detect AF pathology [23–25]. The AA and HBD signals are derived from ECG signals. In Ref. [26], authors have evaluated wavelet domain α -order entropy features from HBD signals to detect paroxysmal AF. Similarly, Park et al. [27] have used a Poincaré plot-based, non-linear signal processing approach to detect AF using HBD signals. Moreover, Nguyen et al. [28] have extracted statistical, geometrical, frequency-domain, Poincaré plot, Lorentz entropy, and hybrid features from HBD signals and used support vector machine (SVM) classifier for the detection of AF. In another work, Cerutti et al. [29] have computed corrected conditional entropy (CCE) and auto-regressive (AR) model-based features from HBD signals for the detection of AF. Similarly, Tateno and Glass [30] have computed the features from the histograms of HBD and Δ HBD signals. They have considered Kolmogorov-Smirnov (KS) test for the classification of normal and AF episodes. Athif et al. [31] have extracted both statistical and morphological features from ECG signals and used SVM classifier to detect AF pathology. Moreover, in Ref. [32], authors have formulated a two-stage signal decomposition framework to detect AF using ECG signals. In the first stage of the decomposition, AA signals have been extracted from the ECG signals. These AA signals are further

decomposed into components in the second stage of the decomposition. The sample entropy features are extracted from the modes of AA, and ECG signals are used to detect AF using a deep belief network (DBN) classifier. Martis et al. [33] have applied independent component analysis (ICA) on ECG beats to extract the features. They have considered Bayesian classifier and ICA-based features from the ECG signal to detect AF. Similarly, Ghosh et al. [15] have computed fractional norm features from the sub-bands of ECG signals. These sub-bands are evaluated using multirate cosine filter-bank-based analysis of ECG signal. They have used a hierarchical extreme learning machine (HELM) based classification approach to detect AF using fractional norm features. The studies mentioned above are based on feature engineering, which requires various processing steps to detect AF using ECG signals.

Recently, various deep learning-based approaches have been applied to detect AF using ECG signals [34]. Andersen et al. [35] have considered a hybrid framework by combining convolutional neural network (CNN) and recurrent neural network (RNN) based models to detect AF using HBD signal. Similarly, Howe and Tizhoosh [36] have used the windowed Fourier transform-based time-frequency representation of ECG signals and a CNN to detect AF. Tran et al. [37] have developed a multi-fusion network-based deep learning approach to detect AF using ECG signals. They have considered different long-short term memory (LSTM) layers and dense layers to construct the multi-fusion network. Shen et al. [38] have compared the classification performance of various machine learning and CNN-based models to detect AF using ECG signals. In recent times, the time-frequency domain deep learning methods have been proposed for the automated analysis of various physiological signals [39,40]. In the time-frequency domain deep learning technique, biomedical signals in the time domain are transformed to joint time-frequency domain signals using different time-frequency-based methods. Then, deep learning methods are used in the time-frequency domain of the physiological signals for classification [39,40]. The chirplet transform has been considered for the time-frequency analysis of various physiological signals [41,42]. It produces high-resolution time-frequency representation compared to the wavelet transform (WT) and windowed Fourier transform methods used to analyze non-stationary signals [42]. The chirplet transform domain time-frequency representation of ECG signal and deep learning techniques have not been explored for the automated detection of AF pathology. The LSTM model exploits temporal correlations of the time-series data in the forward direction [43]. The advantages of bi-directional LSTM (BLSTM) network are: (a) It can exploit the temporal correlations in the time-series data in both forward and backward directions. (b) For periodic signals or time-series data, the BLSTM provides better feature representation as compared to the LSTM [43]. The combination of CNN and BLSTM can exploit spatial and temporal correlations in the time-frequency domain of the ECG signal. The novelty of this study is to develop a deep learning model based on the combination of CNN and BLSTM to detect AF episodes using the time-frequency representation of ECG signals. The major contributions of this paper are given as follows.

- The chirplet transform is used to evaluate the time-frequency domain representation of the ECG signal.
- The time-frequency image of the ECG signal coupled with the deep neural network is used to detect AF and classify terminating vs. non-terminating AF episodes.
- The deep neural network model comprises four convolution layers, one BLSTM layer, and three fully connected layers.
- The proposed deep neural network model is evaluated using ECG signals obtained from three different publicly available databases.

The remaining parts of this paper are structured as follows. The AF pathology-based ECG databases used for the proposed model are described in Section 2. The proposed approach is described in Section 3. The results computed using the proposed deep learning technique for AF

detection and classification of terminating and non-terminating AF episodes are presented in Section 4. The summary of this paper is given in Section 5.

2. ECG databases for AF detection and classification

In this work, the ECG signals were obtained from three public databases to develop our proposed model. The first database is Physionet Computing in Cardiology Challenge 2017, consists of 8482 single-lead ECG recordings divided among AF, normal, and other classes [44,45]. These recordings have a sampling frequency of 300 Hz. The length of ECG recordings varies from 9 s to 60 s, with a median recording length of 30 s [44]. The database consists of 5154 normal recordings, 771 AF recordings, and 2557 records in the ‘other rhythms’ classes. The second database is the mixture of the ECG recordings from MIT-BIH AF dataset [45,46] and MIT-BIH arrhythmia dataset [45,47]. The MIT-BIH AF dataset contains twenty-five records of two-lead ECGs, with each record, has a duration of 60 min. The sampling rate of the dataset is 250 Hz. Similarly, the MIT-BIH arrhythmia dataset consists of forty-seven two-lead ECG recordings, with each 30 min duration and sampled at 360 Hz. This dataset contains ECG signals of normal heart rhythms as well as various other pathological conditions. This study used only normal sinus rhythm (NSR) or healthy control ECG recordings from the MIT-BIH arrhythmia dataset. The third database used in this work is the computing in Cardiology Challenge 2004 AF termination challenge database [6,45]. It consists of thirty records of two-channel ECG signals with the sampling frequency of each is 128 Hz. The recordings are 1 min long and split evenly between non-terminating, terminating after 1 min, and immediately terminating AF classes. Each class contains ten ECG recordings. This study considers the classification schemes such as normal vs. AF and terminating AF vs. non-terminating AF using the ECG recordings from the databases mentioned earlier.

3. Method

The block diagram of time-frequency domain deep learning approach for the detection and classification of AF episodes is shown in Fig. 1. It comprises of four steps: (a) filtering and segmentation of ECG recording into frames, (b) time-frequency analysis of each ECG frame using chirplet transform (c) time-frequency image formulation, and (d) 2D deep convolutional BLSTM for the detection and classification of AF episodes. The brief description of each step of the block-diagram is described in the following sections such as section 2.1, section 2.2, section 2.3 and section 2.4, respectively.

3.1. Filtering and segmentation of ECG recording

To eliminate motion artifacts from the ECG recordings, a Butterworth high-pass filter with a cut-off frequency of 0.5 Hz has been considered in this study [48]. We have also applied a notch filter to eliminate the power-line interference noise from the ECG signals of the physionet AF challenge database [49]. Each ECG recording has been segmented into frames by considering a non-overlapping window of 4-sec duration. We have selected the frame-based segmentation approach to effectively capture the inter-sample and intra-beat correlations in ECG signal analysis [50]. The number of frames extracted from the ECG recordings of each database is shown in Table 1. The NSR, terminating AF, and non-terminating AF episodes of 4 s duration are depicted in Fig. 2 (a), Fig. 2 (b), and Fig. 2 (c), respectively. From these plots, we have observed that there are significant changes in the average RR-interval, QRS-complex morphology, and P-wave shape of both NSR and AF classes. The variations in the QRS-complex morphology and appearance of abnormal atrial activity complexes are the symptoms of AF pathology [21,51]. These pathological variations in the ECG signals can be used to detect AF and classify terminating vs. non-terminating AF episodes. The ECG signal is a non-stationary signal and during AF

pathology, the abnormal atrial activity patterns appear in this signal which is non-linear and time-varying in nature [52]. Therefore, the time-frequency analysis based methods will be helpful to effectively capture the diagnostic information of the ECG signal for the accurate automated identification of AF pathology.

3.2. Chirplet transform for TFR of ECG

The chirplet transform is a method used to get the time-frequency domain representation of non-stationary signals [41]. It has been used in various biomedical applications such as heart valve pathology detection using PCG signals [42], and EEG signal processing [53]. In this study, we have used chirplet transform to compute the time-frequency domain representations of both normal and AF ECG signals. For an ECG signal, $x(n)$, the chirplet transform for computing time-frequency matrix is given as follows [42]:

$$W_{\beta,\sigma}(l, \tau) = \sum_{n=1}^N \tilde{x}(n) e^{-\frac{j2\pi nl}{N}} \Phi_{\tau,\beta,\sigma}^*(n) \quad (1)$$

where, $\tilde{x}(n)$ is the analytic ECG signal obtained using the Hilbert transform of $x(n)$. Here, l and τ are interpreted as frequency and time, respectively. The factor $\Phi_{\tau,\beta,\sigma}^*(n)$ is interpreted as the window function used in chirplet transform, and is defined as follows [41,42]:

$$\Phi_{\tau,\beta,\sigma}^*(n) = G_{\sigma}(n - \tau) e^{-\frac{j\beta}{2}(n - \tau)^2} \quad (2)$$

where, $G_{\sigma}(n)$ is the Gaussian and it is given as follows [42]:

$$G_{\sigma}(n) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{n^2}{2\sigma^2}} \quad (3)$$

In this work, the time-frequency domain representation of both normal and AF ECG signals is evaluated using equation (1). The time-frequency domain representation in equation (1) contains complex values and therefore, we have used the magnitude of $W_{\beta,\sigma}(l, \tau)$ in this work.

The time-frequency domain plots of NSR, terminating AF and non-terminating AF ECG frames are depicted in Fig. 2 (b), Fig. 2 (d), and Fig. 2 (f), respectively. We have seen that the morphologies of time-frequency domain plots are different for NSR, terminating AF, and non-terminating AF frames. In the ECG signal, the variations in the RR-interval, presence of abnormal atrial activity are the symptoms used to diagnose the episodes [45]. The time-frequency plots capture both temporal and frequency information of ECG signals. Hence, the deep learning model developed using both frequency and time-information of ECG signal can help to detect and classify AF automatically. Moreover, other time-frequency analysis methods such as short time Fourier transform (STFT) [54], discrete-time continuous wavelet transform (DT-CWT) [55], and Stockwell transform (ST) [56] have also been used in this work. The classification performance of each time-frequency domain representation of ECG signal is evaluated using deep convolutional BLSTM Network for the detection of AF.

3.3. Deep convolutional BLSTM network

In this work, a novel deep convolutional BLSTM network architecture is proposed for the detection and classification of AF episodes. The Python codes for the deep convolutional BLSTM network is available in the following link². For MIT-BIH and physionet challenge databases, separate deep convolutional BLSTM models are used to classify AF and NSR classes based on the time-frequency images of ECG signals. Similarly, for the classification of non-terminating AF vs. terminating AF episodes, we have also considered a separate deep convolutional BLSTM

² https://github.com/tripathy12345/AF_paper_code.

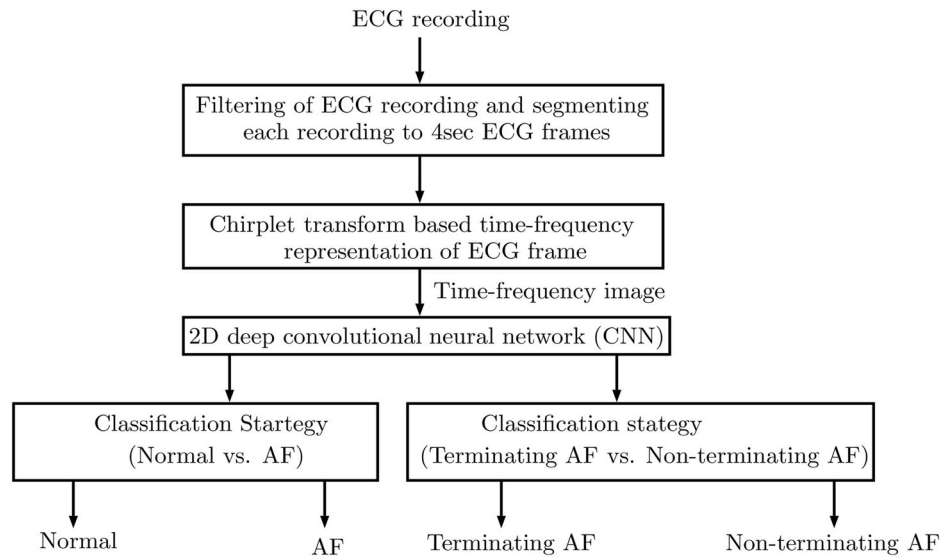


Fig. 1. Block diagram showing the proposed time-frequency domain deep neural network approach to detect AF and classify terminating vs. non-terminating AF episodes.

Table 1

ECG episodes used in this work for the detection and classification of AF.

Database	AF Class	Normal Class	Total
Physionet Challenge database	2180	15077	17257
MIT-BIH AF database	9600	6899	16499
Physionet AF Termination database	Terminating AF 300	Non-terminating AF 150	Total 450

model. The proposed deep neural network model comprises of ten layers and it is shown in Fig. 3. The first and last layers in the network are interpreted as the input and output layers of the proposed deep neural network. The network also consists of four convolution layers, one BLSTM layer, and three fully-connected layers. The time-frequency image of the ECG signal are fed as input to the proposed model. The size of the time-frequency image obtained using the chirplet transform

of ECG signal is 75×1200 . In this study, separate time-frequency domain deep convolutional BLSTM models are trained for three databases. To select the training, and test instances of deep convolutional BLSTM model, we have considered two types of cross-validation (CV) methods: (i) hold-out CV and (ii) 10-fold CV [18]. Since the number of records in each class varies, weighted oversampling is used during training to prevent imbalanced class prediction in the model [18]. We have considered 70%, 20%, and 10% of ECG frames as training, testing and validation for each database using the proposed network with hold-out CV. The mathematical expression used to obtain the two-dimensional (2D) feature-maps (FMs) in the 2D convolution layer of the proposed deep convolutional LSTM network is given as follows [57]:

$$\mathbf{g}_s^l(x, y) = h \left[\sum_{i=1}^T \sum_{j=1}^M \sum_{k=1}^K \mathbf{g}^{l-1}(x+i-1, y+j-1, k) \mathbf{W}^s(i, j, k) + b^s \right] \quad (4)$$

where $\mathbf{W}^s$ is the kernel for sth FM and its size as $t \times m \times K$. K is termed as the number of channels. The parameters T , and M are the size of the

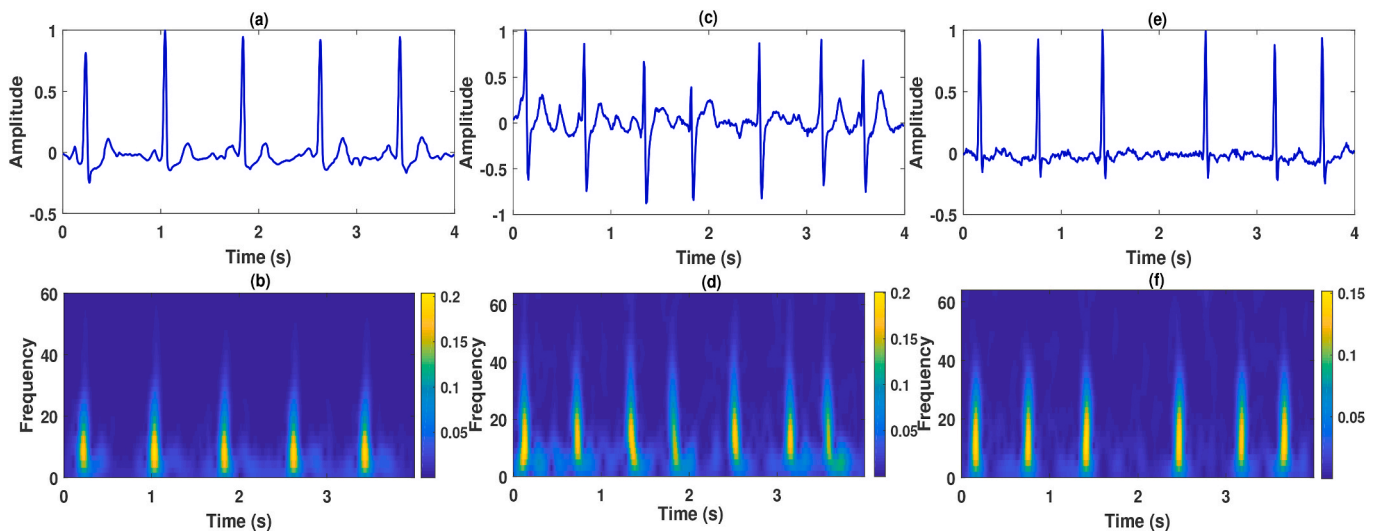


Fig. 2. (a) ECG signal for normal class. (b) Time-frequency plot of normal ECG signal evaluated using chirplet transform. (c) ECG signal for terminating AF class. (d) Time-frequency plot of terminating AF ECG signal evaluated using chirplet transform. (e) ECG signal for non-terminating AF class. (f) Time-frequency domain plot of non-terminating AF ECG signal evaluated using chirplet transform.

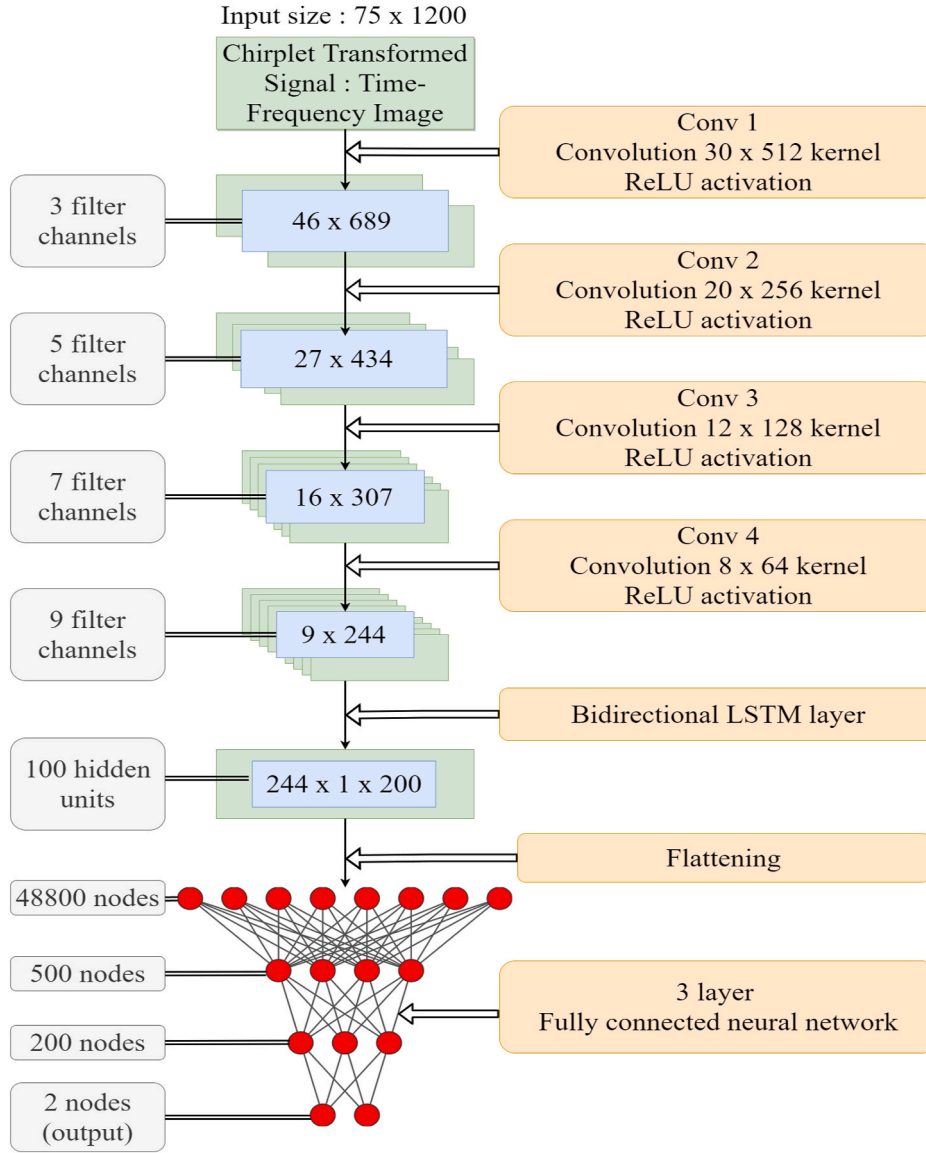


Fig. 3. Proposed time-frequency image based 2D deep convolutional BLSTM network for AF detection and classification.

time-frequency image in the horizontal and vertical directions. Similarly, t , and m are the size of the kernel or filter used in the convolution layer. The $\mathbf{g}^{l-1}$ is the input to the convolution layer. The parameter, $\mathbf{g}_s^l$ is the s th FM evaluated in the l th convolution layer of the proposed deep layer convolutional BLSTM network. The parameter $h[\cdot]$ is denoted as the activation function [18]. After four convolution layers, we have considered a BLSTM layer, and this layer contains 100 neurons. For this BLSTM layer, there are 244 time-steps. The BLSTM cell evaluates the current time-step output by considering the past and future time-step information. The architecture of BLSTM cell mainly comprises of memory cell, forget gate, output gate, update gate and candidate value, respectively for both forward and backward directions. In this work, we have denoted the memory cell for forward and backward directions of BLSTM cell for t th time-step as $\overrightarrow{m}(t)$, and $\overleftarrow{m}(t)$, respectively. Similarly, the mathematical notations for the (forget gate, update gate, output gate, candidate value and output) of BLSTM for forward and backward directions of t th time-step are given as $(\overrightarrow{f}(t), \overrightarrow{u}(t), \overrightarrow{o}(t), \overrightarrow{c}(t), \text{ and } \overrightarrow{z}(t))$, and $(\overleftarrow{f}(t), \overleftarrow{u}(t), \overleftarrow{o}(t), \overleftarrow{c}(t), \text{ and } \overleftarrow{z}(t))$, respectively. The mathematical equations used to evaluate the activation's for forget gate for both forward and backward directions are given as follows [58,59]:

$$\overrightarrow{f}(t) = h(\overrightarrow{\mathbf{W}}_f \cdot [z(t-1), g(t)] + \overrightarrow{\mathbf{b}}_f) \quad (5)$$

$$\overleftarrow{f}(t) = h(\overleftarrow{\mathbf{W}}_f \cdot [z(t+1), g(t)] + \overleftarrow{\mathbf{b}}_f) \quad (6)$$

where $g(t)$, $z(t-1)$, and $z(t+1)$ are the input at t th time-step, output activation at $t-1$ th time-step, and output activation at $t+1$ th time-step, respectively. Similarly, $(\overrightarrow{\mathbf{W}}_f, \text{ and } \overrightarrow{\mathbf{b}}_f)$, and $(\overleftarrow{\mathbf{W}}_f, \text{ and } \overleftarrow{\mathbf{b}}_f)$ are the (weight and bias) at forget gate for forward and backward directions of BLSTM cell. The mathematical equations of update gate for both forward and backward directions are evaluated as follows [58,59]:

$$\overrightarrow{u}(t) = h(\overrightarrow{\mathbf{W}}_u \cdot [z(t-1), g(t)] + \overrightarrow{\mathbf{b}}_u) \quad (7)$$

$$\overleftarrow{u}(t) = h(\overleftarrow{\mathbf{W}}_u \cdot [z(t+1), g(t)] + \overleftarrow{\mathbf{b}}_u) \quad (8)$$

where, $(\overrightarrow{\mathbf{W}}_u, \text{ and } \overrightarrow{\mathbf{b}}_u)$, and $(\overleftarrow{\mathbf{W}}_u, \text{ and } \overleftarrow{\mathbf{b}}_u)$ are the (weight and bias) at update gate for forward and backward directions of BLSTM cell. Moreover, the mathematical expressions for the candidate values and mem-

ory cells for both forward ($\overrightarrow{c(t)}$, and $\overrightarrow{m(t)}$) and backward ($\overleftarrow{c(t)}$, and $\overleftarrow{m(t)}$) directions are evaluated as follows [58,59]:

$$\overrightarrow{c(t)} = \tanh(\overrightarrow{W}_c \cdot [z(t-1), g(t)] + \overrightarrow{b}_c) \quad (9)$$

$$\overrightarrow{m(t)} = \overrightarrow{f(t)} \otimes \overrightarrow{m(t-1)} + \overrightarrow{u(t)} \otimes \overrightarrow{c(t)} \quad (10)$$

$$\overleftarrow{c(t)} = \tanh(\overleftarrow{W}_c \cdot [z(t+1), g(t)] + \overleftarrow{b}_c) \quad (11)$$

$$\overleftarrow{m(t)} = \overleftarrow{f(t)} \otimes \overleftarrow{m(t+1)} + \overleftarrow{u(t)} \otimes \overleftarrow{c(t)} \quad (12)$$

where ($\overrightarrow{W}_c, \overrightarrow{b}_c$), and ($\overleftarrow{W}_c, \overleftarrow{b}_c$) are the (weight, bias) to obtain candidate value for forward and backward directions of BLSTM. Similarly, the mathematical equations of output gate and final output activation in forward ($\overrightarrow{o(t)}$, and $\overrightarrow{z(t)}$) and backward ($\overleftarrow{o(t)}$, and $\overleftarrow{z(t)}$) directions for t th time-step are computed as follows [58,59]:

$$\overrightarrow{o(t)} = f(\overrightarrow{W}_o \cdot [z(t-1), g(t)] + \overrightarrow{b}_o) \quad (13)$$

$$\overrightarrow{z(t)} = \overrightarrow{o(t)} \otimes \tanh(\overrightarrow{m(t)}) \quad (14)$$

$$\overleftarrow{o(t)} = f(\overleftarrow{W}_o \cdot [z(t+1), g(t)] + \overleftarrow{b}_o) \quad (15)$$

$$\overleftarrow{z(t)} = \overleftarrow{o(t)} \otimes \tanh(\overleftarrow{m(t)}) \quad (16)$$

where ($\overrightarrow{W}_o, \overrightarrow{b}_o$), and ($\overleftarrow{W}_o, \overleftarrow{b}_o$) are the (weight, bias) of the output gate for forward and backward directions of BLSTM [58]. In this work, BLSTM layer in the proposed deep neural network model (as shown in Fig. 3) contains 100 hidden neurons. Therefore, the feature map produced at the output of BLSTM layer has the size as $244 \times 1 \times 200$. Moreover, for dense or fully-connected layer, the feature vectors at l th layer is computed as follows [57]:

$$\mathbf{z}^l = g(\mathbf{V}^l \mathbf{z}^{l-1} + \mathbf{b}^l) \quad (17)$$

where $\mathbf{V}$ is the weight matrix between $l-1$ th dense layer and l th dense layer, respectively. Similarly, $\mathbf{z}^l, \mathbf{z}^{l-1}$, and $\mathbf{b}^l$ are the feature vector (FV) at l th dense layer, FV at $l-1$ th dense layer, and bias value at l th dense layer, respectively. The training parameters of the proposed deep learning model is shown in Table 2. The performance measures such as accuracy, F1-score, Mathew's correlation coefficient (MCC), sensitivity and specificity are used to measure the performance of the proposed deep neural network based approach [18,60,61].

4. Results and discussion

The results evaluated for detection and classification of AF episodes using time-frequency-based 2D CNN are presented in this section. First, we have shown the obtained results with hold-out CV using three databases. In Fig. 4(a) and Fig. 4(b), we have shown the accuracy vs. epoch graphs evaluated for training and validation of time-frequency-based 2D deep convolutional BLSTM network using ECG signals from Physionet challenge and MIT-BIH databases, respectively. Moreover, the accuracy vs. epoch graphs computed for training and validation instances using the proposed deep neural network model for terminating vs. non-terminating AF classification scheme is shown in Fig. 4(c). The accuracy value increased with the increase of epochs in the plots of normal vs. AF and terminating AF vs. non-terminating AF classification strategies. For ECG signals obtained from the MIT-BIH database, both validation and training accuracy values of the proposed time-frequency-based 2D deep convolutional BLSTM network are more than 98% for NSR vs. AF classification strategy. Similarly, training and validation

accuracy values of 2D deep convolutional BLSTM network are more than 90% using ECG signals obtained from Physionet challenge AF database. Moreover, for terminating AF vs. non-terminating AF classification strategy, the training accuracy value is more than 95% after 200 epochs. However, the validation accuracy varied between 75% and 85%. This study has computed the results by considering five random trial-based hold-out CV schemes [18]. The average and standard deviation values obtained by considering all five random trial-based hold-out CVs are shown. The confusion matrix evaluated for one trial of hold-out CV using the ECG signals taken from MIT-BIH, physionet challenge 2017, and computing in Cardiology Challenge 2004 AF termination challenge databases are shown in Fig. 5, Fig. 6, and Fig. 7, respectively. It can be observed from the confusion matrix plots that the number of false negatives and false positives are high using the ECG signals obtained from the Physionet challenge 2017 database compared to the MIT-BIH database. The accuracy values obtained using the proposed approach for one trial of hold-out CV using MIT-BIH database and Physionet challenge AFDB ECG signals are 98.8% and 96%, respectively. Similarly, for the non-terminating AF vs. terminating AF classification scheme, the accuracy obtained using the proposed time-frequency-based 2D convolutional BLSTM approach for one trial of the hold-out CV is 74%. The classification results obtained for the test ECG instances using the proposed time-frequency-based 2D convolutional BLSTM network for Physionet challenge, MIT-BIH AF, and Physionet AF termination databases with hold-out based CV strategy are depicted in Table 3. It can be observed that the time-frequency-based 2D deep convolutional BLSTM network has obtained the classification metrics such as sensitivity, accuracy, and specificity values of more than 98% by considering the ECG signals from the MIT-BIH database for normal vs. AF classification scheme. Similarly, the MCC and F1-score values are also more than 97%. Moreover, the time-frequency-based 2D deep convolutional BLSTM network has obtained the sensitivity, accuracy, and specificity values of 89.76%, 91.74%, and 95.75% by considering the ECG signals from the Physionet AF challenge database for normal vs. AF classification strategy. Similarly, for terminating AF vs. non-terminating AF classification strategy, the time-frequency-based 2D deep convolutional BLSTM network has demonstrated sensitivity, accuracy, specificity, and F1-score values of 74.55%, 74.29%, 74.17%, and 64.08%, respectively.

The classification results obtained for the proposed time-frequency domain 2D convolutional BLSTM using ECG signals obtained from Physionet challenge AF database and 10-fold CV scheme are shown in Table 4. It can be observed that the accuracy values are more than 90% for each fold of NSR and AF the classification. Similarly, for fold 1 and fold 7, the sensitivity values of time-frequency domain deep convolutional BLSTM are 88.34%. The sensitivity values are obtained for other folds as more than 90% using the proposed deep learning-based model. The MCC and F1-score values are found to be more than 70% in each fold. The 10-fold CV-based classification results of the time-frequency domain deep convolutional BLSTM model obtained using ECG signals from MIT-BIH AF database are shown in Table 5. It can be observed that the accuracy, sensitivity, and specificity values are more than 98% for each fold. Moreover, the average F1-score and average MCC values evaluated by considering all 10-folds are 99.17% and 98%, respectively, using our proposed method. The classification results obtained using the proposed time-frequency domain deep convolutional BLSTM for terminating AF vs. non-terminating AF classification scheme with 10-fold CV are shown in Table 6. It can be seen that the accuracy, sensitivity, and specificity values of the time-frequency-based 2D deep convolutional BLSTM are more than 70% for each fold. In fold 7 and fold 8, the accuracy values of 81.43% and 80%, respectively, are obtained using the deep learning model. The average F1-score and MCC values of 66.02% and 48.58% are obtained using the proposed deep learning model for terminating AF vs. non-terminating AF classification scheme. From both 10-fold CV and hold-out CV results, we have also observed that the proposed time-frequency domain-based deep convolutional BLSTM has demonstrated higher classification performance using the ECG signals

Table 2

Details of hyper-parameter values used for our proposed work.

Database	Learning rate	Regularisation	Frame size	batch size	Epochs	Optimizer
Physionet Challenge database	10^{-5}	0.01	1200	32	60	Adam
MIT-BIH AF database	10^{-7}	0.003	1000	32	30	Adam
Physionet AF Termination database	10^{-4}	0.1	512	64	200	Adam

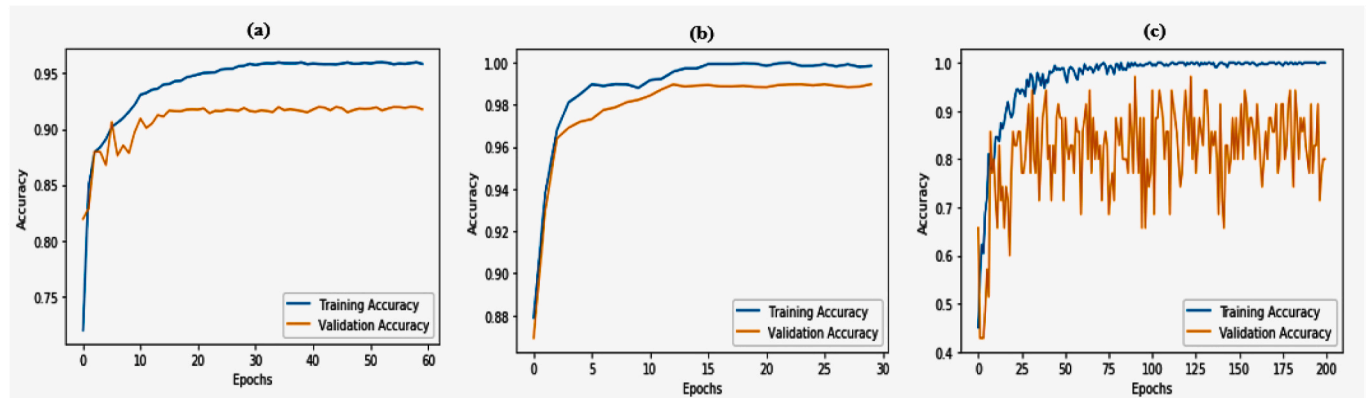


Fig. 4. (a) Accuracy vs. epoch graph of time-frequency based deep convolutional BLSTM for normal vs. AF classification methodology using ECG signals from physionet AF challenge database. (b) Accuracy vs. epoch graph of time-frequency based deep convolutional BLSTM for normal vs. AF classification methodology using ECG signals from MIT-BIH AF database. (c) Accuracy vs. epoch plot of time-frequency based deep CNN for terminating AF vs. non-terminating AF classification scheme using ECG signals from physionet AF termination database.

obtained from the MIT-BIH database than the ECG signals considered from the physionet challenge AF database. The physionet challenge AF database ECG signals contain noise, and this may be why the performance of the proposed time-frequency domain-based deep convolutional BLSTM is less for this database. Moreover, the terminating AF and non-terminating AF episodes are highly correlated with the variations in the ECG signal [9,62]. For this reason, the proposed approach has demonstrated less average accuracy value than NSR vs. AF classification

schemes using ECG signals obtained from both physionet challenge and MIT-BIH databases for both types of CV schemes.

Moreover, we have evaluated the classification performance of deep convolutional BLSTM network using other time-frequency analysis approaches such as short time-Fourier transform (STFT), discrete-time continuous wavelet transform (DT-CWT), and Stockwell transform (ST), respectively. The classification results for each of the time-frequency analysis methods with convolutional BLSTM are shown in

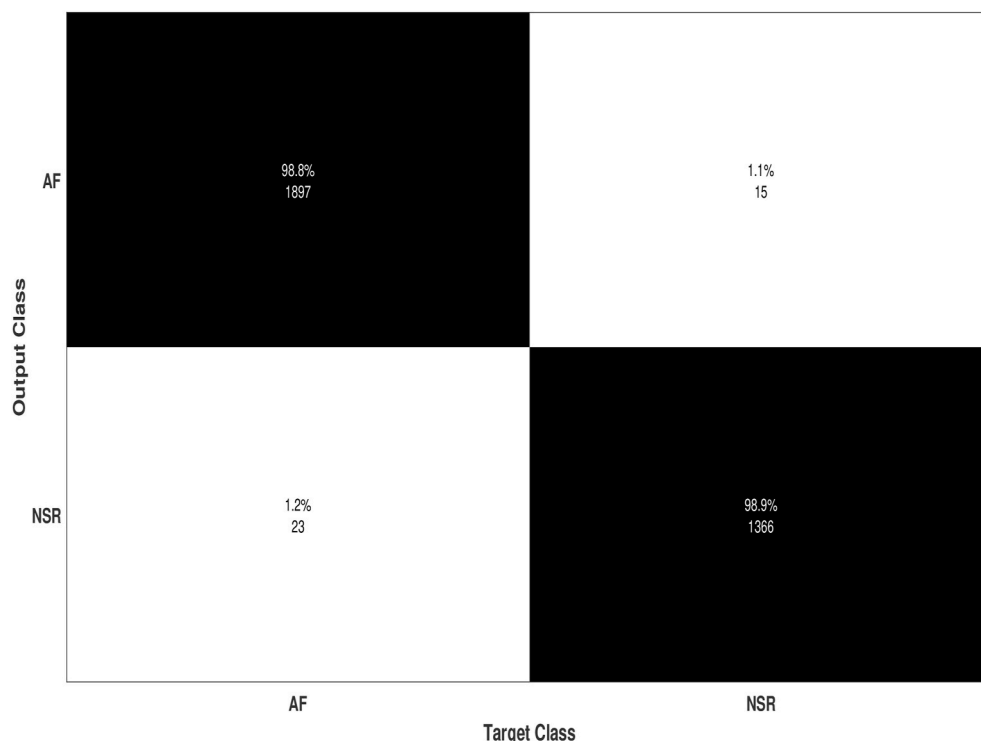


Fig. 5. Confusion matrix obtained for one-trial of hold-out CV using our proposed approach with ECG signals obtained from MIT-BIH AFDB database.

Table 7. It can be observed that the STFT and DT-CWT-based time-frequency analysis methods combined with deep convolutional BLSTM models have obtained average accuracy values of 97.82%, and 97.90%, respectively. The convolutional BLSTM coupled with ST-based time-frequency images of ECG signals have produced the average accuracy value as 89.60% which is less than STFT and DT-CWT-based methods. The chirplet transform-based time-frequency images combined with convolutional BLSTM have obtained higher classification results than ST, DT-CWT, and STFT based time-frequency analysis techniques. The Kruskal-Wallis (KW) based statistical test [63,64] is performed to verify the statistical significance in the differences in the classification performance of deep convolutional BLSTM classifier using different time-frequency analysis methods. The p-values computed for accuracy, F-score, MCC, and specificity measures using the KW test are 0.007, 0.004, 0.012, and 0.007, respectively. For sensitivity measure, the p-value obtained using the ANOVA test is 0.080. The significant differences in the mean values of accuracy, F-score, MCC, and specificity measures are observed for convolutional BLSTM classifiers using various time-frequency domain approaches. The differences in the mean sensitivity values are less for deep convolutional BLSTM classifiers with STFT, DTCWT, ST, and CT-based time-frequency representations of ECG signals. Hence, the p-value of sensitivity measure is greater than 0.05.

We have evaluated the classification performance of the convolutional BLSTM network using the chirplet transform-based time-frequency images of both 5 s and 8-sec frames of ECG signals. The results are shown in Table 8. It is evident that the time-frequency images of 5 s and 8 s ECG frames coupled with convolutional BLSTM models have obtained the average accuracy values of 97.20%, and 96.65%, respectively. These accuracy values are less than the accuracy value of convolutional BLSTM obtained using time-frequency images 4 s ECG frames. The other performance measures such as sensitivity, specificity, F1-score, and MCC values obtained using convolutional BLSTM for the time-frequency images of 5 s and 8 s ECG frames are less as those of the time-frequency images of 4 s ECG frames. Hence, we have selected 4 s based segmentation of ECG recordings in this work to detect AF. Moreover, the KW test is also used to verify the statistical differences in

the classification results of the convolutional BLSTM classifier using the time-frequency images of different ECG durations. The p-values obtained for accuracy, specificity, F1-score measures are 0.047, 0.009, and 0.017, respectively. Similarly, the p-values obtained for sensitivity, and MCC measures are 0.136 and 0.084, respectively. The $p < 0.05$ for accuracy measure reveals that the differences in the classification accuracy values of convolutional BLSTM classifiers are significant using the time-frequency images of 4 s, 5 s, and 8 s ECG durations.

Moreover, we have compared the classification performance of the proposed time-frequency domain convolutional BLSTM approach with various machine learning and deep learning based techniques developed for detecting AF automatically using both MIT-BIH database and physionet challenge database, respectively. In Table 9, we have shown the comparison of the proposed approach with existing machine learning based methods using ECG signals from MIT-BIH database for AF detection. In Ref. [30], authors have considered histograms of the HBD signal, first derivative of HBD signal (ΔHBD) and KS test-based probabilistic classifier model to detect AF episodes. They have obtained the specificity and sensitivity values of 97.20%, and 94.40% respectively for their work. Similarly, the RMS of successive RR interval differences and Shannon entropy features have been used in Ref. [65] to detect AF. Their work has obtained the specificity and sensitivity values of 95.10%, and 94.40%, respectively using a threshold-based binary classification approach. The variational mode decomposition (VMD) domain entropy features of ECG signals have been used by Tripathy et al [51]. to detect AF. They have obtained the specificity and sensitivity values of 98.87%, and 97.67%, respectively, using DBN based deep-learning classifier. The flexible analytic wavelet transform (FAWT) domain entropy features coupled with random forest-based model have demonstrated the specificity and sensitivity values of 97.60% and 95.80%, respectively [66]. Moreover, the approach based on the evaluation of features using the wavelet domain correlation matrix of ECG signal and artificial neural network (ANN) model has obtained the specificity and sensitivity values of 98.90% and 98.70%, for the AF detection [67]. The features calculated based on time-varying coherence function of ECG signal coupled with threshold-based binary classifier have yielded the specificity and

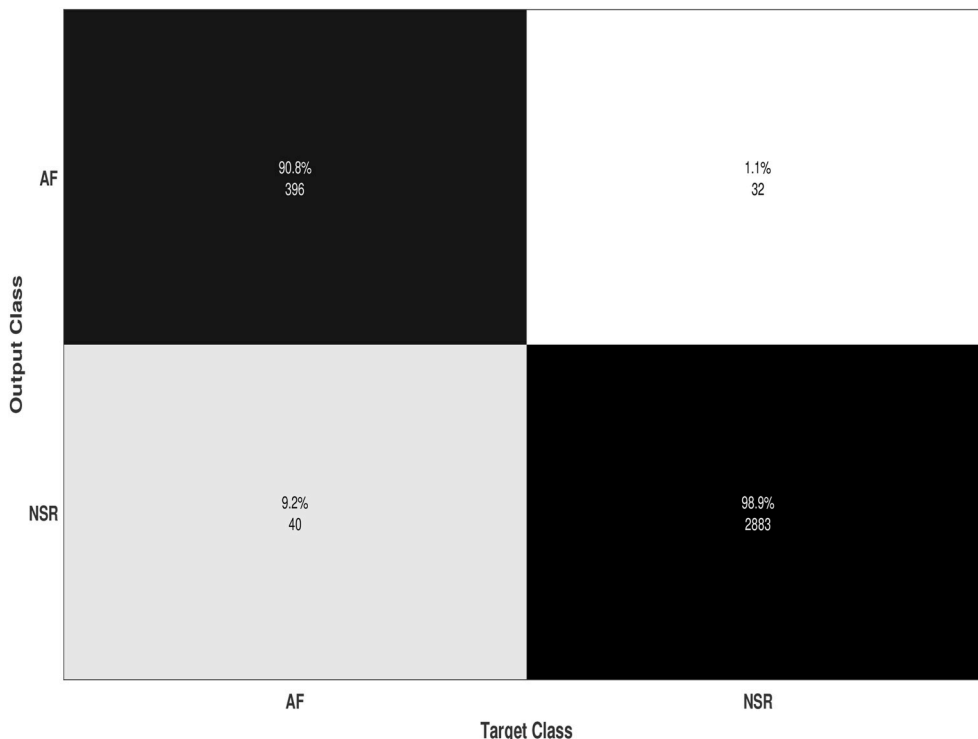


Fig. 6. Confusion matrix obtained for one-trial of hold-out CV using our proposed approach with ECG signals obtained from Physionet challenge 2017.

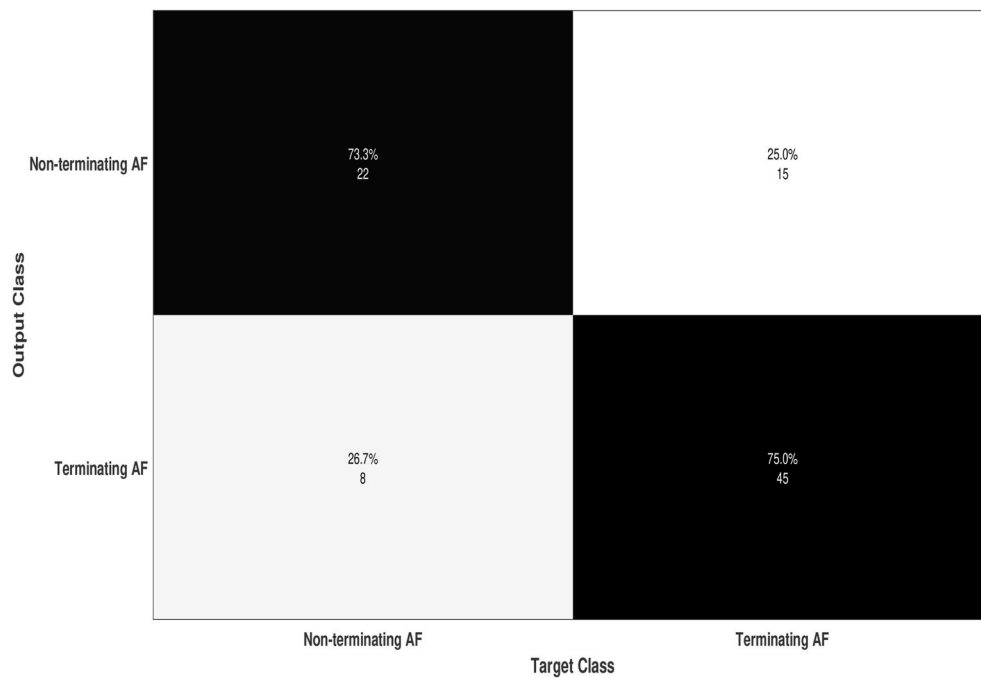


Fig. 7. Confusion matrix obtained for one-trial of the hold-out CV using proposed learning approach for non-terminating AF vs. terminating AF classification scheme.

Table 3

Classification results obtained using the proposed approach with hold-out CV technique.

Database	Classification Scheme	Accuracy (%)	F1-score	MCC	Sensitivity (%)	Specificity (%)
Physionet Challenge database	Normal vs. AF	91.74 ± 1.15	0.749 ± 0.025	0.722 ± 0.027	89.76 ± 1.16	95.75 ± 1.16
MIT-BIH AF database	Normal vs. AF	98.92 ± 0.61	99.24 ± 0.43	97.38 ± 1.46	98.91 ± 0.62	98.94 ± 0.57
Physionet AF Termination database	Terminating AF vs. Non-Terminating AF	74.29 ± 3.91	64.08 ± 4.42	45.36 ± 7.25	74.55 ± 4.07	74.17 ± 4.32

Table 4

Performance obtained for AF detection using our proposed method in each folds of the 10-fold (FD) CV using from Physionet Challenge database ECG signals.

Measure	FD 1	FD 2	FD 3	FD 4	FD 5	FD 6	FD 7	FD 8	FD 9	FD 10	Average ($\mu \pm \sigma$)
Accuracy (%)	90.26	93.03	92.89	93.34	93.31	93.40	90.26	93.34	92.23	92.74	92.48 ± 1.23
F1-score	72.52	78.97	78.52	79.69	79.33	79.79	72.74	79.44	76.95	78.03	77.60 ± 2.75
MCC	70.40	77.24	76.68	77.95	77.32	78.02	70.82	77.46	75.04	76.04	75.70 ± 2.83
Sensitivity (%)	88.34	90.93	90.93	91.36	91.36	91.36	88.34	91.36	90.28	90.71	90.50 ± 1.19
Specificity (%)	94.27	97.04	96.87	97.33	97.30	97.40	94.27	97.33	96.21	96.74	96.48 ± 1.22

Table 5

Performance obtained for AF detection using our proposed method in each folds of the 10-fold (FD) CV using MIT-BIH database ECG signals.

Measure	FD 1	FD 2	FD 3	FD 4	FD 5	FD 6	FD 7	FD 8	FD 9	FD 10	Average ($\mu \pm \sigma$)
Accuracy (%)	99.37	99.57	98.37	99.11	99.69	98.60	99.09	98.94	99.40	99.63	99.18 ± 0.44
F1-score	99.56	99.70	98.85	99.38	99.78	99.01	99.36	99.26	99.58	99.74	99.42 ± 0.31
MCC	98.46	98.95	96.06	97.83	99.23	96.62	97.78	97.45	98.53	99.09	98.00 ± 1.06
Sensitivity (%)	99.36	99.56	98.36	99.12	99.68	98.60	99.08	98.96	99.40	99.64	99.17 ± 0.44
Specificity (%)	99.40	99.60	98.40	99.10	99.70	98.60	99.10	98.90	99.40	99.60	99.18 ± 0.42

Table 6

Performance obtained for automated classification of terminating AF vs. non-terminating AF ECG signals using our proposed method in each folds of the 10-fold (FD) CV using Physionet Challenge database.

Measure	FD1	FD 2	FD 3	FD 4	FD 5	FD 6	FD 7	FD 8	FD 9	FD 10	Average ($\mu \pm \sigma$)
Accuracy (%)	71.43	74.29	77.14	75.71	78.57	71.43	81.43	80.00	74.29	74.29	75.86 ± 3.40
F1-score	61.54	65.38	65.22	65.31	70.59	60.00	73.47	69.57	62.50	66.67	66.02 ± 4.21
MCC	40.86	47.08	48.35	47.50	55.51	38.95	60.15	54.83	43.49	49.07	48.58 ± 6.69
Sensitivity (%)	72.73	77.27	77.27	77.27	77.27	68.18	81.82	81.82	72.73	77.27	76.36 ± 4.18
Specificity (%)	70.83	72.92	77.08	75.00	79.17	72.92	81.25	79.17	75.00	71.43	75.48 ± 3.57

Table 7

Classification results obtained using STFT, DT-CWT, and ST based convolutional BLSTM classifier using ECG signals from MIT-BIH database for hold-out CV.

Time-frequency analysis method	Accuracy (%)	Sensitivity	Specificity	F1-score (%)	MCC (%)
STFT	97.82 ± 1.28	97.50 ± 0.50	97.46 ± 1.69	97.93 ± 0.99	97.10 ± 0.89
DT-CWT	97.90 ± 1.63	97.72 ± 1.87	97.62 ± 2.06	97.87 ± 1.65	97.29 ± 1.33
ST	89.60 ± 4.37	98.36 ± 1.04	82.40 ± 6.60	90.24 ± 3.91	81.29 ± 7.35

Table 8

Classification results obtained using chirplet based time-frequency domain convolutional BLSTM for different duration ECG signals from MIT-BIH database.

Window size	Accuracy (%)	Sensitivity	Specificity	F1-score (%)	MCC (%)
5 s	97.20 ± 1.52	98.52 ± 1.08	94.67 ± 3.12	97.24 ± 1.66	94.56 ± 2.87
8 s	96.65 ± 2.01	97.93 ± 3.74	93.89 ± 3.64	96.77 ± 1.89	93.55 ± 3.74

sensitivity values of 97.91%, and 97.67%, respectively, in detecting the AF episodes automatically [68]. It can be observed from the comparison results that the proposed time-frequency domain deep convolutional BLSTM approach has obtained higher classification performance than the existing feature-engineering-based techniques developed to detect AF episodes.

The performance of the proposed time-frequency domain deep convolutional BLSTM is also compared with various deep learning-based methods in Table 10 using MIT-BIH database ECG signals. The CNN [70] and multiscale fusion-based deep CNN [69] based deep neural network models have obtained the sensitivity values of 90.20%, and 92.41%, respectively, and these values are less than the sensitivity obtained using the proposed approach. The specificity values of these deep neural network-based methods (shown in Table 10) are also less than the specificity value obtained using our proposed approach. Similarly, the multiscale convolutional attention-based bidirectional gated recurrence unit (BGRU) model has demonstrated the specificity and sensitivity values of 97.50%, and 97.80%, respectively, for the categorization of AF and NSR classes using ECG signals [71]. The HAN-ECG-based deep-learning approach has produced the specificity and sensitivity values of 98.54% and 99.08%, respectively, in detecting AF using ECG signals obtained from the MIT-BIH database. Similarly, the STFT based time-frequency image of ECG signal coupled with deep CNN model has obtained the sensitivity and specificity values of 98.34%, and 98.24%, respectively [72]. The same authors in Ref. [72] have also shown that the combination of stationary wavelet transform (SWT) based time-frequency image of ECG signal and deep CNN produced the sensitivity and specificity values of 98.79%, and 97.87%, respectively. The modified frequency slice wavelet transform (MFSWT) based time-frequency image of ECG signal and deep CNN has produced the sensitivity and specificity values of 79.05%, and 89.99%, respectively. The proposed approach has obtained higher specificity and sensitivity values than all the deep learning methods in Table 10 to detect AF episodes using ECG signals. We have used the KW test [63,64] to evaluate the statistical differences in the AF detection sensitivity and specificity values of various deep learning approaches shown in Table 10. The p-values obtained for sensitivity and specificity measures using KW test are 2.70×10^{-13} , and 1.93×10^{-11} , respectively. It has been observed that the differences in both sensitivity and specificity values of the

proposed approach and other deep learning methods are significant with p-values less than 0.05. We have evaluated the 95% confidence interval (CI) of sensitivity and specificity for each deep learning method to detect AF using ECG signals. These values are shown in Table 10, and it may be noted that, for each deep learning method, average sensitivity and average specificity values appear in the CI ranges. It may be noted that there are overlapping in the CI ranges for the proposed approach (method 8) vs. method 6, and method 8 vs. method 4. For overlapping CI ranges, the *t*-test has been used to verify the statistical difference in the samples of different groups [4]. We have applied the *t*-test using the sensitivity values of method 6 vs. method 8, and method 4 vs. method 8. The p-values of sensitivity measures obtained for method 6 vs. method 8, and method 4 vs. method 8 using *t*-test are less than 0.05. The lower p-values reveal that the differences in the sensitivity values between the deep learning methods are statistically significant for detecting AF. It is observed that there is no overlapping between the CI range of specificity measure of the proposed approach with existing deep learning-based methods. However, there are overlapping in the CI ranges of method 1 vs. method 3, method 3 vs. method 5, method 2 vs. method 6, and method 3 vs. method 6 (see Table 10).

In Table 11, we have shown the summary of various methods which have been reported for automated AF detection using ECG signals with same Physionet challenge AF database. The features extracted from HBD and atrial activity signals coupled with neural network classifier have obtained the specificity and sensitivity values of 95.50%, and 19.92%, respectively [74]. Similarly, the CNN-based approach proposed by Pourbabaee et al. [70] have evaluated the specificity and sensitivity values of 93.31%, and 59.21%, respectively. Similarly, the HAN-ECG approach proposed by Mousavi et al. [58] have obtained a sensitivity value of 86.02%. The proposed approach has obtained higher classification performance than the results obtained by the methods in Ref. [74], and [70], respectively. The sensitivity of the proposed approach is higher than the HAN-ECG-based AF detection approach using ECG data from the Physionet challenge AF database. The multiscale fusion based CNN approach has evaluated the sensitivity and specificity values of 92.41% and 97.70% for the detection of AF episodes using Physionet AF challenge database ECG signals. Our proposed time-frequency domain deep learning approach has obtained slightly less classification performance compared to the multiscale fusion based CNN. The computational

Table 9

Summary of the machine learning based methods used for automated AF detection using ECG signals with same MIT-BIH AFDB database.

Method used	Sensitivity (%)	Specificity (%)
Histograms of HBD and Δ HBD as features and Kolmogorov-Smirnov (KS) test based classifier [30]	94.40	97.20
Shannon entropy and root mean square (RMS) of successive RR interval differences and threshold classifier [65]	94.40	95.10
Entropy features extracted in the VMD domain of ECG signal and deep belief network (DBN) [51]	97.67	98.87
Entropy features evaluated in the FAWT of ECG data and random forest classifier [66]	95.80	97.60
Features computed from the wavelet domain correlation matrix of ECG signal and artificial neural network (ANN) classifier [67]	98.70	98.90
Time-varying coherence function based features extracted from ECG signal and threshold classifier [68]	97.67	97.91
Time-frequency domain convolutional BLSTM (Proposed work)	99.17	99.18

Table 10

Summary of deep learning based methods used for automated AF detection using ECG signals with same MIT-BIH AFDB database.

Sl. no	Method used	Sensitivity (%)	Specificity (%)	CI sensitivity	CI specificity
1	Multiscale fusion based deep CNN model [69]	92.41	97.70	[91.30, 93.52]	[97.40, 98.01]
2	CNN based learnable feature extraction from ECG and classification [70]	90.20	90.48	[89.32, 91.09]	[89.45, 91.49]
3	Multiscale convolutional attention based bidirectional gated recurrent unit (BGRU) model [71]	97.80	97.50	[97.67, 97.93]	[96.67, 98.31]
4	Hierarchical attention network (HAN) based on wave attention, beat attention and window attention [58]	99.08	98.54	[98.79, 99.37]	[98.36, 98.71]
5	STFT based time-frequency image of ECG signal and deep CNN [72]	98.34	98.24	[98.25, 98.43]	[97.92, 98.54]
6	SWT based time-frequency image of ECG signal and deep CNN [72]	98.79	97.87	[98.70, 98.88]	[97.76, 97.97]
7	Modified frequency slice wavelet transform (MFSWT) and deep CNN [73]	79.05	89.99	[76.49, 82.51]	[88.57, 91.40]
8	Time-frequency domain convolutional BLSTM (Proposed work)	99.17	99.18	[98.85, 99.49]	[98.86, 99.49]

complexity of the proposed deep learning model to detect AF in the testing phase is evaluated as follows. The deep convolutional BLSTM network consists of four convolution layers, one BLSTM layer, and three FC layers. The computational complexity of all four convolution layers is evaluated as $O(\sum_{l=1}^4 T^{(l)} M^{(l)} t^{(l)} m^{(l)} I^{(l)} S^{(l)})$ [75]. where ' $T^{(l)}$ ' and ' $M^{(l)}$ ' are the spatial dimensions of the input to the l th convolution layer. The parameters ' $t^{(l)}$ ' and ' $m^{(l)}$ ' are the spatial dimensions of the kernel or filter for l th convolution layer. Similarly, the parameters ' $I^{(l)}$ ' and ' $S^{(l)}$ ' are termed as the number of input and output feature maps for l th convolution layer. The computational complexity of BLSTM layer is evaluated as $O(2U)$ [76] [77] [78]. where $U = 4AB + 4B^2 + 3B + BC$. where A and C are the number of inputs and outputs of BLSTM. B is the number of cells in the hidden layer of BLSTM. The computational complexity evaluated by considering all three FC layers is given as $O(\sum_{l=1}^3 q^{(l)} r^{(l)} s^{(l)} \log(s^{(l)}))$ [75]. where $q^{(l)}$ and $r^{(l)}$ are the size of weight matrix in the l th FC layer. The parameter $s^{(l)}$ is the number of output neurons in the l th FC layer. Thus, the total computational complexity of the proposed deep convolutional BLSTM model is evaluated as $O(\sum_{l=1}^4 T^{(l)} M^{(l)} t^{(l)} m^{(l)} I^{(l)} S^{(l)} + O(2U) + O(\sum_{l=1}^3 q^{(l)} r^{(l)} s^{(l)} \log(s^{(l)}))$. The multiscale fusion based CNN simultaneously process two-channel ECG signals for the detection of AF and it consists of 26 convolution layers (for each channel 13 convolution layers), 10 max pooling layers (for each channel 5 pooling layers), and 3 FC layers, respectively. The computational complexity of a single pooling layer is $O(F^{(l)})$. where $F^{(l)}$ is the size of the input to the l th pooling layer. The total computational complexity of multiscale fusion based CNN model is evaluated as $O(\sum_{l=1}^{26} T^{(l)} M^{(l)} t^{(l)} m^{(l)} I^{(l)} S^{(l)} + O(\sum_{l=1}^{26} F^{(l)}) + O(\sum_{l=1}^3 q^{(l)} r^{(l)} s^{(l)} \log(s^{(l)}))$. From the computational complexity expressions in terms of 'O' notations, it is observed that the proposed convolutional BLSTM network has less complexity as that of the multiscale fusion based CNN model for AF detection. The software program for the detection of AF using the time-frequency images of ECG frames is written in the python framework. The google co-lab with the specifications as NVIDIA Tesla T4 GPU, 12.72 GB RAM have been used to simulate the python program and evaluate results of the proposed AF detection, and classification approach [79]. The training time for the proposed deep convolutional BLSTM model using time-frequency images of 5 s ECG frames is 960 s or 16 min. The testing time evaluated for the deep learning model is 0.045 s.

The novelty of our developed time-frequency domain deep learning model is that we are not only able to detect AF with highest accuracy (comparable with [69]), also our system is able to classify terminating and non-terminating AF classes with more than 70% accuracy. The advantages of the proposed approach are given below:

- A novel convolutional BLSTM based deep learning model is proposed for automated AF detection using time-frequency images of ECG signals.
- Developed system can decipher the subtle temporal and spatial correlations of ECG signals in the time-frequency plane.
- Demonstrated highest classification performance to detect AF using ECG signals.
- Suggested approach is computationally faster as compared to the multiscale fusion-based deep CNN model for AF detection.
- Generated model is also able to classify terminating AF vs. non-terminating AF classes automatically. To the best of our knowledge, we are the first group to perform this classification in addition to automated AF detection using the time-frequency domain deep learning technique.

The two-stage classification schemes, such as normal vs. AF in stage 1 and non-terminating AF vs. terminating AF in stage 2, are interesting for diagnosing AF pathology using ECG signals. In the future, the two-stage deep learning models can be developed to detect and classify AF episodes. The real-time implementations of proposed AF detection and classification approaches can be performed to design internet of things (IoT)-based intelligent cardiovascular healthcare systems. In this work, we have used only 20 ECG records from the computing in Cardiology Challenge 2004 AF termination challenge database to classify terminating AF vs. non-terminating AF episodes. The classification performance can be further improved using more ECG recordings of both terminating AF and non-terminating AF classes. In this work, we have considered only the chirplet transform-based time-frequency method to analyze ECG signals. The spline kernel-based chirplet transform has obtained better time-frequency resolution for the heart sound signals [40]. The spline chirplet transform of ECG signal coupled with deep learning can improve the classification performance with a vast database.

Table 11

Summary of methods used for automated AF detection using ECG signals with same Physionet challenge 2017 database.

Method used	Sensitivity (%)	Specificity (%)
Features computed from Atrial activity and HBD signals and neural network classifier [74]	19.92	95.50
CNN-based deep learning approach [70]	59.21	93.31
Multiscale fusion based CNN approach [69]	92.41	97.70
Hierarchical attention network based on wave attention, beat attention and window attention [58]	86.02	98.62
Time-frequency domain convolutional BLSTM (Proposed work)	90.50	96.48

5. Conclusion

A novel time-frequency domain deep convolutional BLSTM network is proposed to detect AF and classify terminating vs. non-terminating AF episodes automatically using ECG signals. The chirplet transform-based time-frequency analysis method has been applied to evaluate time-frequency images from the ECG signals. The convolutional BLSTM classifier model is used to detect AF pathology using time-frequency images of ECG signals. The proposed approach has obtained an accuracy of more than 90% and 99% for physionet challenge and MIT-BIH databases, respectively. The deep convolutional BLSTM model has demonstrated higher classification performance using chirplet transform-based time-frequency representation of ECG signal as compared to STFT, DT-CWT, and ST-based methods. Moreover, the approach has obtained an accuracy value higher than 70% for the classification of terminating vs. non-terminating AF episodes. In the future, we intend to validate our model with more diverse AF databases. Due to the superior performance, the proposed time-frequency domain deep learning model can be deployed in intelligent healthcare systems for automated AF detection using ECG sensor data.

Declaration of competing interest

This is an academic research work. I declare that there is no conflict of interest for this paper.

Acknowledgments

Dr. Rajesh Kumar Tripathy has received the funding through an OPERA Grant, BITS Pilani, India for this work. The grant number for this funded research project is BITS/GAU/RIG/2019/H0632.

References

- [1] J. Mackay, G.A. Mensah, The Atlas of Heart Disease and Stroke, World Health Organization, 2004.
- [2] S. Nattel, New ideas about atrial fibrillation 50 years on, *Nature* 415 (6868) (2002) 219–226.
- [3] R. Wächter, M. Weber-Krüger, J. Seegers, F. Edelmann, J. Wohlfahrt, K. Wasser, G. Gelbrich, G. Hasenfuß, R. Stahrenberg, J. Liman, et al., Age-dependent yield of screening for undetected atrial fibrillation in stroke patients: the find-af study, *J. Neurol.* 260 (8) (2013) 2042–2045.
- [4] C. van Walraven, R.G. Hart, S. Connolly, P.C. Austin, J. Mant, F.R. Hobbs, P. J. Koudstaal, P. Petersen, F. Perez-Gomez, J.A. Kottnerus, et al., Effect of age on stroke prevention therapy in patients with atrial fibrillation: the atrial fibrillation investigators, *Stroke* 40 (4) (2009) 1410–1416.
- [5] A. Matarese, C. Sardu, J. Shu, G. Santulli, Why is chronic obstructive pulmonary disease linked to atrial fibrillation? a systematic overview of the underlying mechanisms, *Int. J. Cardiol.* 276 (2019) 149.
- [6] G. Moody, Spontaneous termination of atrial fibrillation: a challenge from physionet and computers in cardiology 2004, *Computers in Cardiology*, 2004 (2004) 101–104. IEEE.
- [7] G.K. Moe, On the multiple wavelet hypothesis of atrial fibrillation, *Arch. Int. Pharmacodyn. Ther.* 140 (1962) 183.
- [8] M.A. Allesie, Experimental evaluation of moe's multiple wavelet hypothesis of atrial fibrillation, *Cardiac electrophysiology and arrhythmias* (1985) 265–275.
- [9] R. Alcaraz, F. Sandberg, L. Sornmo, J.J. Rieta, Classification of paroxysmal and persistent atrial fibrillation in ambulatory ecg recordings, *IEEE (Inst. Electr. Electron. Eng.) Trans. Biomed. Eng.* 58 (5) (2011) 1441–1449.
- [10] R. Sun, Y. Wang, Predicting spontaneous termination of atrial fibrillation based on the rr interval, *Proc. IME H J. Eng. Med.* 223 (6) (2009) 713–726.
- [11] C.T. January, L.S. Wann, J.S. Alpert, H. Calkins, J.E. Cigarroa, J.C. Cleveland, J. B. Conti, P.T. Ellinor, M.D. Ezekowitz, M.E. Field, et al., 2014 aha/acc/hrs guideline for the management of patients with atrial fibrillation: a report of the american college of cardiology/american heart association task force on practice guidelines and the heart rhythm society, *J. Am. Coll. Cardiol.* 64 (21) (2014) e1–e76.
- [12] T.-S. Kim, H.-J. Youn, Role of echocardiography in atrial fibrillation, *J. Cardiovascul. Ultrasound* 19 (2) (2011) 51.
- [13] K. Harris, D. Edwards, J. Mant, How can we best detect atrial fibrillation? *J. Roy. Coll. Phys. Edinb.* 42 (2012) 5–22.
- [14] S.Z. Rosero, V. Kutiyfa, B. Olshansky, W. Zareba, Ambulatory ecg monitoring in atrial fibrillation management, *Prog. Cardiovasc. Dis.* 56 (2) (2013) 143–152.
- [15] S.K. Ghosh, R.K. Tripathy, M.R. Paternina, J.J. Arrieta, A. Zamora-Mendez, G. R. Naik, Detection of atrial fibrillation from single lead ecg signal using multirate cosine filter bank and deep neural network, *J. Med. Syst.* 44 (2020) 1–15.
- [16] H. Habibzadeh, K. Dinesh, O.R. Shishvan, A. Boggio-Dandry, G. Sharma, T. Soyata, A survey of healthcare internet of things (hiot): a clinical perspective, *IEEE Internet of Things J.* 7 (1) (2019) 53–71.
- [17] J. Lee, B.A. Reyes, D.D. McManus, O. Maitas, K.H. Chon, Atrial fibrillation detection using an iphone 4s, *IEEE (Inst. Electr. Electron. Eng.) Trans. Biomed. Eng.* 60 (1) (2012) 203–206.
- [18] R. Panda, S. Jain, R. Tripathy, U.R. Acharya, Detection of shockable ventricular cardiac arrhythmias from ecg signals using ffewt filter-bank and deep convolutional neural network, *Comput. Biol. Med.* 124 (2020) 103939.
- [19] O. Faust, M. Kareem, A. Shenfield, A. Ali, U.R. Acharya, Validating the robustness of an internet of things based atrial fibrillation detection system, *Pattern Recogn. Lett.* 133 (2020) 55–61.
- [20] S. Ladavich, B. Ghorani, Rate-independent detection of atrial fibrillation by statistical modeling of atrial activity, *Biomed. Signal Process Contr.* 18 (2015) 274–281.
- [21] A. Chetan, R.K. Tripathy, S. Dandapat, A diagnostic system for detection of atrial and ventricular arrhythmia episodes from electrocardiogram, *J. Med. Biol. Eng.* 38 (2) (2018) 304–315.
- [22] U. Maji, M. Mitra, S. Pal, Automatic detection of atrial fibrillation using empirical mode decomposition and statistical approach, *Proced. Technol.* 10 (2013) 45–52.
- [23] Y. Hagiwara, H. Fujita, S.L. Oh, J.H. Tan, R. San Tan, E.J. Ciaccio, U.R. Acharya, Computer-aided diagnosis of atrial fibrillation based on ecg signals: a review, *Inf. Sci.* 467 (2018) 99–114.
- [24] U.R. Acharya, O. Faust, E.J. Ciaccio, J.E.W. Koh, S.L. Oh, R. San Tan, H. Garan, Application of nonlinear methods to discriminate fractionated electrograms in paroxysmal versus persistent atrial fibrillation, *Comput. Methods Progr. Biomed.* 175 (2019) 163–178.
- [25] O. Faust, A. Shenfield, M. Kareem, T.R. San, H. Fujita, U.R. Acharya, Automated detection of atrial fibrillation using long short-term memory network with rr interval signals, *Comput. Biol. Med.* 102 (2018) 327–335.
- [26] Y. Xin, Y. Zhao, Paroxysmal atrial fibrillation recognition based on multi-scale wavelet α -entropy, *Biomed. Eng. Online* 16 (1) (2017) 1–12.
- [27] J. Park, S. Lee, M. Jeon, Atrial fibrillation detection by heart rate variability in poicare plot, *Biomed. Eng. Online* 8 (1) (2009) 1–12.
- [28] A. Nguyen, S. Ansari, M. Hooshmand, K. Lin, H. Ghanbari, J. Gryak, K. Najarian, Comparative study on heart rate variability analysis for atrial fibrillation detection in short single-lead ecg recordings, in: 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), IEEE, 2018, pp. 526–529.
- [29] S. Cerutti, L. Mainardi, A. Porta, A. Bianchi, Analysis of the dynamics of rr interval series for the detection of atrial fibrillation episodes, *Computers in Cardiology* 1997 (1997) 77–80. IEEE.
- [30] K. Tateno, L. Glass, Automatic detection of atrial fibrillation using the coefficient of variation and density histograms of rr and δ rr intervals, *Med. Biol. Eng. Comput.* 39 (6) (2001) 664–671.
- [31] M. Athif, P.C. Yasawardene, C. Daluwatte, Detecting atrial fibrillation from short single lead ecgs using statistical and morphological features, *Physiol. Meas.* 39 (6) (2018), 064002.
- [32] R.K. Tripathy, M.R. Paternina, J.G. Arrieta, P. Pattanaik, Automated detection of atrial fibrillation ecg signals using two stage vmd and atrial fibrillation diagnosis index, *J. Mech. Med. Biol.* 17 (2017) 1740044, 07.
- [33] R.J. Martis, U.R. Acharya, H. Prasad, C.K. Chua, C.M. Lim, Automated detection of atrial fibrillation using bayesian paradigm, *Knowl. Base Syst.* 54 (2013) 269–275.
- [34] Z. Ebrahimi, M. Loni, M. Daneshmand, A. Gharehbaghi, A review on deep learning methods for ecg arrhythmia classification, *Expert Syst. Appl.* X (2020) 100033.
- [35] R.S. Andersen, A. Peimankar, S. Puthusserypady, A deep learning approach for real-time detection of atrial fibrillation, *Expert Syst. Appl.* 115 (2019) 465–473.
- [36] S. Ross-Howe, H.R. Tizhoosh, Atrial fibrillation detection using deep features and convolutional networks, in: 2019 IEEE EMBS International Conference on Biomedical & Health Informatics (BHI), IEEE, 2019, pp. 1–4.
- [37] L. Tran, Y. Li, L. Nocera, C. Shahabi, L. Xiong, Multifusionnet: atrial fibrillation detection with deep neural networks, *AMIA Summits on Translational Science Proceedings* 2020 (2020) 654.
- [38] M. Shen, L. Zhang, X. Luo, J. Xu, Atrial fibrillation detection algorithm based on manual extraction features and automatic extraction features, in: IOP Conference Series: Earth and Environmental Science, vol. 428, IOP Publishing, 2020, 012050.
- [39] S. Madhavan, R.K. Tripathy, R.B. Pachori, Time-frequency domain deep convolutional neural network for the classification of focal and non-focal ecg signals, *IEEE Sensor. J.* 20 (6) (2019) 3078–3086.
- [40] S.K. Ghosh, R. Ponnalagu, R. Tripathy, U.R. Acharya, Deep layer kernel sparse representation network for the detection of heart valve ailments from the time-frequency representation of pcg recordings, *BioMed Res. Int.* 2020 (2020).
- [41] S. Mann, S. Haykin, The chirplet transform: physical considerations, *IEEE Trans. Signal Process.* 43 (11) (1995) 2745–2761.
- [42] S.K. Ghosh, R. Ponnalagu, R. Tripathy, U.R. Acharya, Automated detection of heart valve diseases using chirplet transform and multiclass composite classifier with pcg signals, *Comput. Biol. Med.* 118 (2020) 103632.
- [43] S. Jeong, I. Park, H.S. Kim, C.H. Song, H.K. Kim, Temperature prediction based on bidirectional long short-term memory and convolutional neural network combining observed and numerical forecast data, *Sensors* 21 (3) (2021) 941.
- [44] A. Physionet, Classification from a short single lead ecg recording: the physionet/computing in cardiology challenge, 2017.
- [45] A.L. Goldberger, L.A. Amaral, L. Glass, J.M. Hausdorff, P.C. Ivanov, R.G. Mark, J. E. Mietus, G.B. Moody, C.-K. Peng, H.E. Stanley, PhysioBank, physioToolkit, and physionet: components of a new research resource for complex physiologic signals, *Circulation* 101 (23) (2000) e215–e220.

- [46] G. Moody, A new method for detecting atrial fibrillation using rr intervals, *Comput. Cardiol.* (1983) 227–230.
- [47] G.B. Moody, R.G. Mark, The impact of the mit-bih arrhythmia database, *IEEE Eng. Med. Biol. Mag.* 20 (3) (2001) 45–50.
- [48] L. Sharma, R. Tripathy, S. Dandapat, Multiscale energy and eigenspace approach to detection and localization of myocardial infarction, *IEEE Trans. Biomed. Eng.* 62 (7) (2015) 1827–1837.
- [49] P.S. Hamilton, A comparison of adaptive and nonadaptive filters for reduction of power line interference in the ecg, *IEEE Trans. Biomed. Eng.* 43 (1) (1996) 105–109.
- [50] S. Padhy, S. Dandapat, Exploiting multi-lead electrocardiogram correlations using robust third-order tensor decomposition, *Healthcare technology letters* 2 (5) (2015) 112–117.
- [51] R. Tripathy, M.R.A. Paternina, J.G. Arrieta, P. Pattanaik, Automated detection of atrial fibrillation ecg signals using two stage vmd and atrial fibrillation diagnosis index, *J. Mech. Med. Biol.* 17 (2017) 1740044, 07.
- [52] T. Tsalaile, R. Sameni, S. Sanei, C. Jutten, J. Chambers, et al., Sequential blind source extraction for quasi-periodic signals with time-varying period, *IEEE (Inst. Electr. Electron. Eng.) Trans. Biomed. Eng.* 56 (3) (2008) 646–655.
- [53] J. Cui, W. Wong, The adaptive chirplet transform and visual evoked potentials, *IEEE (Inst. Electr. Electron. Eng.) Trans. Biomed. Eng.* 53 (7) (2006) 1378–1384.
- [54] A. Ullah, S.M. Anwar, M. Bilal, R.M. Mehmood, Classification of arrhythmia by using deep learning with 2-d ecg spectral image representation, *Rem. Sens.* 12 (10) (2020) 1685.
- [55] R. He, K. Wang, N. Zhao, Y. Liu, Y. Yuan, Q. Li, H. Zhang, Automatic detection of atrial fibrillation based on continuous wavelet transform and 2d convolutional neural networks, *Front. Physiol.* 9 (2018) 1206.
- [56] R.K. Tripathy, M.R. Paternina, J.G. Arrieta, A. Zamora-Méndez, G.R. Naik, Automated detection of congestive heart failure from electrocardiogram signal using stockwell transform and hybrid classification scheme, *Comput. Methods Progr. Biomed.* 173 (2019) 53–65.
- [57] P. Jain, P. Gajbhiye, R. Tripathy, U.R. Acharya, A two-stage deep cnn architecture for the classification of low-risk and high-risk hypertension classes using multi-lead ecg signals, *Informatics in Medicine Unlocked* 21 (2020) 100479.
- [58] S. Mousavi, F. Afghah, U.R. Acharya, Han-ecg: an interpretable atrial fibrillation detection model using hierarchical attention networks, *Comput. Biol. Med.* 127 (2020) 104057.
- [59] A. Varshney, S.K. Ghosh, S. Padhy, R.K. Tripathy, U.R. Acharya, Automated classification of mental arithmetic tasks using recurrent neural network and entropy features obtained from multi-channel ecg signals, *Electronics* 10 (9) (2021).
- [60] C.M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
- [61] D. Chicco, G. Jurman, The advantages of the matthews correlation coefficient (mcc) over f1 score and accuracy in binary classification evaluation, *BMC Genom.* 21 (1) (2020) 1–13.
- [62] C.B. De Vos, R. Pisters, R. Nieuwlaet, M.H. Prins, R.G. Tieleman, R.-J.S. Coelen, A. C. van den Heijkant, M.A. Allesie, H.J. Crijns, Progression from paroxysmal to persistent atrial fibrillation: clinical correlates and prognosis, *J. Am. Coll. Cardiol.* 55 (8) (2010) 725–731.
- [63] P.E. McKight, J. Najab, Kruskal-wallis test, *The corsini encyclopedia of psychology* (2010), <https://doi.org/10.1002/9780470479216.corpsy0491>, 1–1.
- [64] P. Gajbhiye, N. Mingchinda, W. Chen, S.C. Mukhopadhyay, T. Wilaiprasitporn, R. K. Tripathy, Wavelet domain optimized savitzky-golay filter for the removal of motion artifacts from eeg recordings, *IEEE Transactions on Instrumentation and Measurement* 70 (2020) 1–11.
- [65] S. Dash, K. Chon, S. Lu, E. Raeder, Automatic real time detection of atrial fibrillation, *Ann. Biomed. Eng.* 37 (9) (2009) 1701–1709.
- [66] M. Kumar, R.B. Pachori, U.R. Acharya, Automated diagnosis of atrial fibrillation ecg signals using entropy features extracted from flexible analytic wavelet transform, *Biocybernetics and Biomedical Engineering* 38 (3) (2018) 564–573.
- [67] J. Wang, P. Wang, S. Wang, Automated detection of atrial fibrillation in ecg signals based on wavelet packet transform and correlation function of random process, *Biomed. Signal Process Contr.* 55 (2020) 101662.
- [68] J. Lee, Y. Nam, D.D. McManus, K.H. Chon, Time-varying coherence function for atrial fibrillation detection, *IEEE (Inst. Electr. Electron. Eng.) Trans. Biomed. Eng.* 60 (10) (2013) 2783–2793.
- [69] X. Fan, Q. Yao, Y. Cai, F. Miao, F. Sun, Y. Li, Multiscaled fusion of deep convolutional neural networks for screening atrial fibrillation from single lead short ecg recordings, *IEEE journal of biomedical and health informatics* 22 (6) (2018) 1744–1753.
- [70] B. Pourbabae, M.J. Roshtkhari, K. Khorasani, Deep convolutional neural networks and learning ecg features for screening paroxysmal atrial fibrillation patients, *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 48 (12) (2018) 2095–2104.
- [71] T. Wang, Y. Qin, A novel multi-scale convolutional network with attention-based bidirectional gated recurrent unit for atrial fibrillation discrimination, *Biocybernetics and Biomedical Engineering* 41 (2) (2021) 445–455.
- [72] Y. Xia, N. Wulan, K. Wang, H. Zhang, Detecting atrial fibrillation by deep convolutional neural networks, *Comput. Biol. Med.* 93 (2018) 84–92.
- [73] X. Xu, S. Wei, C. Ma, K. Luo, L. Zhang, C. Liu, Atrial fibrillation beat identification using the combination of modified frequency slice wavelet transform and convolutional neural networks, *Journal of healthcare engineering* 2018 (2018).
- [74] R. Couceiro, P. Carvalho, J. Henriques, M. Antunes, M. Harris, J. Habetha, Detection of atrial fibrillation using model-based ecg analysis, in: 2008 19th International Conference on Pattern Recognition, IEEE, 2008, pp. 1–5.
- [75] C. Ding, S. Liao, Y. Wang, Z. Li, N. Liu, Y. Zhuo, C. Wang, X. Qian, Y. Bai, G. Yuan, et al., Circnn: accelerating and compressing deep neural networks using block-circulant weight matrices, in: Proceedings of the 50th Annual IEEE/ACM International Symposium on Microarchitecture, 2017, pp. 395–408.
- [76] K. Kawakami, Supervised Sequence Labelling with Recurrent Neural Networks, Ph. D. thesis, 2008.
- [77] J. Schmidhuber, S. Hochreiter, et al., Long short-term memory, *Neural Comput.* 9 (8) (1997) 1735–1780.
- [78] U. Lotrič, A. Dobnikar, Parallel implementations of recurrent neural network learning, in: International Conference on Adaptive and Natural Computing Algorithms, Springer, 2009, pp. 99–108.
- [79] E. Bisong, *Building Machine Learning and Deep Learning Models on Google Cloud Platform: A Comprehensive Guide for Beginners*, Apress, 2019.